

RIQICHA AI

Multilingual Recruitment Intelligence Platform

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1. Abstract

Riqicha AI is a recruitment automation system designed for Ethiopia's linguistically diverse job market. It uses NLP to automate CV parsing and matching across Afaan Oromoo, Amharic, Somali, Sidama, Wolayttattuwa, Arabic, and English.

2. Problem Statement

Traditional hiring is hindered by:

- Manual Screening: Overwhelming volume of applicants leads to fatigue.
- Language Barrier: ATS systems lack local language support.
- Bias and Inefficiency: Slow hiring cycles and subjective selection.

3. Proposed Solution

- AI Pipeline: End-to-end automation from upload to ranking.
- Cross-Lingual Matching: Skills mapped across different scripts.
- Voice Integration: Result summaries via AI voice.

4. System Architecture

High-Level Stack

- Frontend: React.js + Tailwind CSS
- Backend: Flask (Python) API
- AI Layer: NLP (Spacy/Transformers)
- Database: MongoDB (NoSQL)

The frontend communicates with a REST API. The backend orchestrates NLP processing, saving document metadata to MongoDB while returning real-time scores to the UI.

5. System Features

Core Capabilities

- CV Document Parsing (PDF/DOCX)
- Multilingual Entity Recognition
- Automated Candidate Ranking
- Accessibility-driven Voice Summaries

6. AI Logic Explanation

1. Extraction: Texts are tokenized using language-specific models.
2. NER: Customized Named Entity Recognition identifies skills and experience.
3. Embeddings: Profiles are converted into numerical vectors.
4. Match Score: Euclidean distance and keyword weighting create a 0-100 score.

7. Database Design (MongoDB)

Collections

- Users: Authentication and role data.
- Candidates: Parsed CV text, extracted skills, and profiles.
- Jobs: Requirements and industry category.

8. System Workflow

1. Upload -> 2. Language ID -> 3. NLP Extraction -> 4. ATS Scoring -> 5. Dashboard Display.

9. Results & Output

ANALYSIS COMPLETE

Candidate: Hana B.

Detected Category: IT/Software

Top Skills: React, Python, MongoDB

Overall Match Score: 92%

10. Future Improvements

- Mobile Native App (Flutter/React Native)
- Real-time Interview Scheduling
- Blockchain-verified Credentials

11. Conclusion

Riqicha AI streamlines recruitment in Ethiopia by bridging the language gap and automating technical screening. It ensures fairness and speed in the national job market.